RETAIL SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI

PROJECT REPORT

Submitted By:

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Tool Used: Microsoft Power BI Desktop

1. Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps me organize business data and present useful information through interactive reports, charts, tables and KPI cards.

For this project, I created a Retail Sales Dashboard using Microsoft Power BI. I used sales, cost, profit, quantity, order, product category, store type, city, store and customer-segment information to review retail performance.

The dashboard brings together key performance indicators and visual comparisons. It helps me examine revenue by product category, store type, month, city and individual store. I also included filters for year, store type, product category, state and customer segment, along with a Store Type Tooltip page.

2. Objectives of the Project

The main objectives of my project are:

- To summarize total revenue, profit, orders and quantity sold.
- To validate important sales and financial measures.
- To compare revenue across product categories and store types.
- To review revenue patterns across months.
- To compare revenue generated by different cities and stores.
- To analyse the dashboard using year, store type, product category, state and customer-segment filters.
- To display additional store-level performance measures through a tooltip page.

3. Dataset Description

I used retail sales data containing information about products, sales revenue, cost, profit, quantity, transactions, orders, stores, store types, cities, states and customer segments. The dashboard uses these fields to compare sales performance across different business dimensions.

The main fields and measures used in my report are:

Field / Measure	Description
Total Quantity	Shows the total quantity of products sold.
Total Revenue	Shows the combined revenue generated from sales.
Total Cost	Displays the total cost associated with the sales.
Total Profit	Shows the difference between total revenue and total cost as represented by the report.
Average Unit Price	Displays the average unit price measure.
Total Transactions / Orders	Shows the transaction and order count measures.
Average Order Value	Displays the average order value measure.
Product Category	Groups sales into Accessories, Audio, Home Appliances, Laptops, Mobile Phones, Televisions and Wearables.
Store Type / Store Name	Supports revenue comparisons across store formats and individual stores.
City / State	Supports geographical sales comparisons and filtering.
Customer Segment	Groups customers into the segments shown in the filters.

4. Data Validation and Summary Measures

I used a DAX Validation page to review the main measures before presenting them in the dashboard. The validation table displays total quantity of 112,353, total revenue of 4,70,70,20,747.02, total cost of 3,72,71,29,600 and total profit of 97,98,91,147.02. It also displays an average unit price of 41,989.0144, 15,000 total transactions, 15,000 total orders and an average order value of approximately 3,13,801.38.

The dashboard rounds these figures for display. For example, total revenue is shown as 4.71bn, total profit as 979.89M, total orders as 15K and total quantity as 112K. The validation page provides the more detailed figures, while the dashboard cards make the main results easier to read.

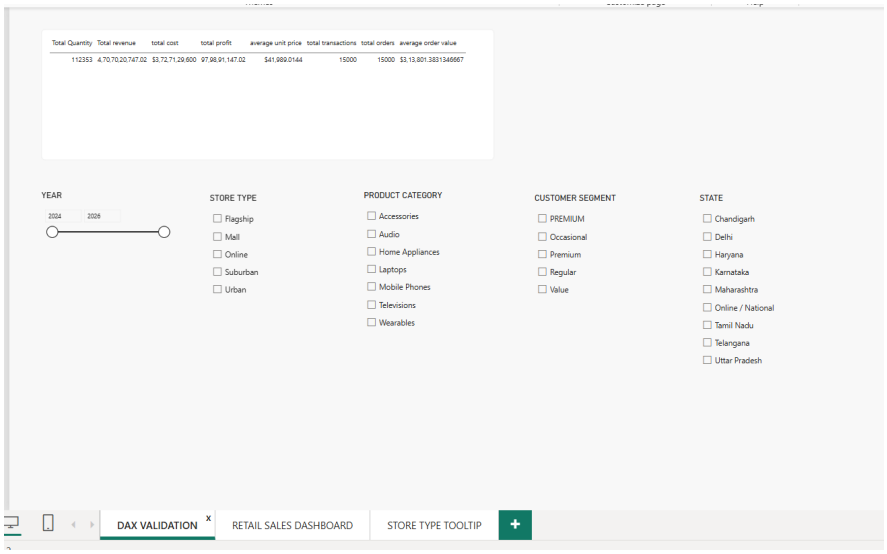


Figure 1: DAX validation table and report filters.

5. KPI Cards

I used four KPI cards at the top of the Retail Sales Dashboard to summarize the main performance measures:

- **Total Revenue:** 4.71bn.
- **Total Profit:** 979.89M.
- **Total Orders:** 15K.
- **Total Quantity:** approximately 112K.

These cards give me a quick overview of sales performance without needing to inspect every chart. The validation table also shows 15,000 total transactions and 15,000 total orders in the displayed data.

6. Dashboard Components

I created several visualizations to compare retail performance across product, store and geographical dimensions.

6.1 Revenue by Product Category

I used a column chart to compare total revenue across seven product categories. Mobile Phones has the highest displayed revenue at approximately 1.77bn, followed by Laptops at 1.46bn. Televisions contributes around 0.93bn. Home Appliances generates approximately 0.31bn, Audio around 0.16bn and Wearables around 0.08bn. Accessories has a very small displayed value, close to 0.00bn at the chart's rounding level.

Purpose: To identify the product categories that contribute most and least to total revenue.

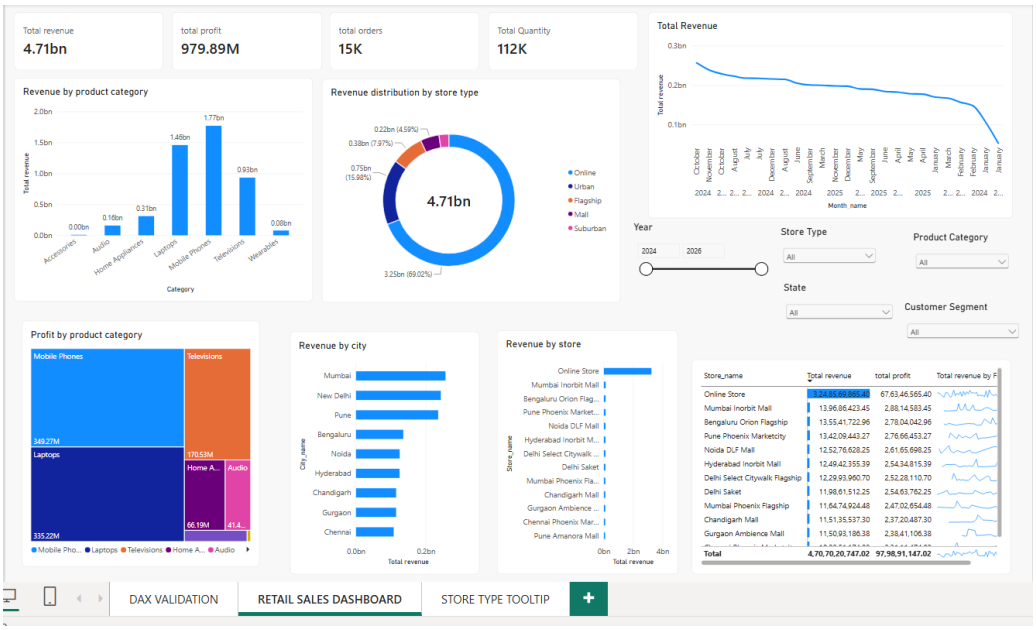


Figure 2: Retail sales dashboard with category, store type, city and store analysis.

6.2 Revenue Distribution by Store Type

I used a doughnut chart to show how total revenue is distributed across store types. Online contributes approximately 3.25bn, or 69.02% of the total. Urban contributes around 0.75bn (15.98%), Flagship around 0.38bn (7.97%) and Mall around 0.22bn (4.59%). Suburban represents a very small remaining share in the chart.

Purpose: To compare the contribution of different store formats to total revenue.

6.3 Total Revenue by Month

I used a line chart to display total revenue across the Month_name field. The visible line begins at approximately 0.25bn and generally moves downward across the displayed sequence, ending near 0.05bn. The axis labels contain repeated month names alongside year labels, and the chart appears to show a descending sequence rather than a simple January-to-December order. I therefore describe the displayed pattern without assuming that the axis is sorted in normal calendar order.

Purpose: To review revenue variation across the displayed month and year combinations.

6.4 Profit by Product Category

I used a treemap to compare profit across product categories. Mobile Phones and Laptops occupy the largest areas, with displayed profits of approximately 349.27M and 335.22M. Televisions follows at around 170.53M. Home Appliances contributes approximately 66.19M, while Audio has a smaller displayed value. The remaining smaller categories occupy much less space in the treemap.

Purpose: To compare the relative profit contribution of each product category.

6.5 Revenue by City

I used a horizontal bar chart to compare revenue across cities. Mumbai has the longest visible bar, followed by New Delhi and Pune. Bengaluru, Noida, Hyderabad, Chandigarh, Gurgaon and Chennai appear below these cities in the displayed chart. This visual helps me compare geographical sales performance.

Purpose: To identify cities with higher displayed revenue and compare their contribution.

6.6 Revenue by Store

I used a horizontal bar chart to compare total revenue across individual stores. Online Store has the longest bar by a wide margin, while the remaining listed stores have much smaller revenue values on the same scale. The adjacent table provides store-level details, including total revenue, total profit and a small revenue trend visualization. The visible table includes stores such as Mumbai Inorbit Mall, Bengaluru Orion Flagship, Pune Phoenix Marketcity and Noida DLF Mall.

Purpose: To compare individual store performance and inspect revenue and profit details at store level.

7. Interactive Filters

I added the following slicers to make the dashboard interactive:

- **Year:** Allows me to select a year within the displayed range of 2024 to 2026.
- **Store Type:** Allows me to filter by Flagship, Mall, Online, Suburban or Urban.
- **Product Category:** Allows me to focus on a selected product category.
- **Customer Segment:** Allows me to filter by the displayed customer segments: Premium, Occasional, Regular and Value.
- **State:** Allows me to focus on the listed states and regions, including Chandigarh, Delhi, Haryana, Karnataka, Maharashtra, Online / National, Tamil Nadu, Telangana and Uttar Pradesh.

These slicers allow me to narrow the report to selected periods, store formats, product categories, customer segments and states. The visuals can then be reviewed under the selected filter conditions.

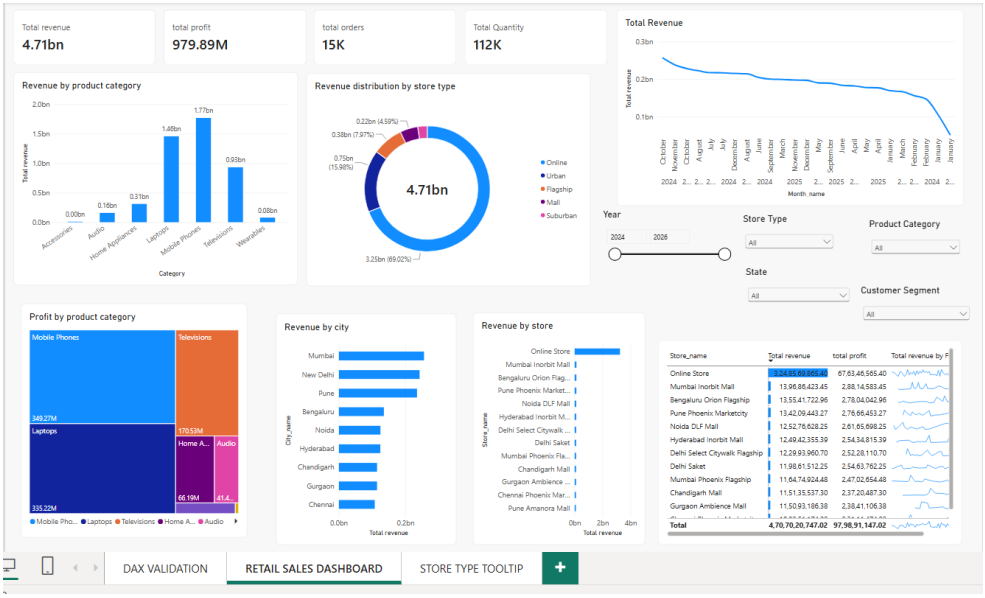


Figure 3: Interactive Retail Sales Dashboard and slicers.

8. Store Type Tooltip

I created a separate Store Type Tooltip page containing four KPI cards: Total Revenue of 4.71bn, Total Profit of 979.89M, Total Orders of 15K and Average Order Value displayed as approximately 313.801... . The average order value is truncated in the card, so I use the detailed validation figure of approximately 3,13,801.38 when describing the measure.

When the tooltip page is assigned to a related visual in Power BI, it can display these summary measures when I hover over a store type or related data point. This gives additional context without leaving the main dashboard.

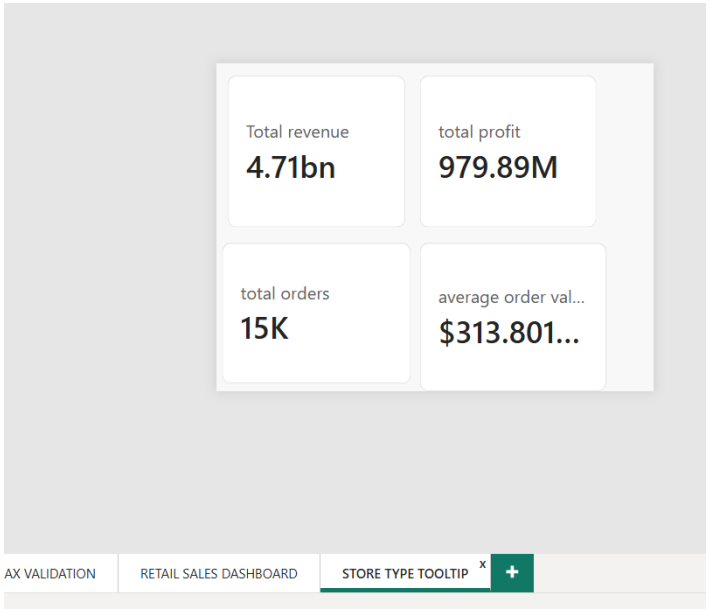


Figure 4: Store Type Tooltip page with revenue, profit, orders and average order value.

9. Key Findings

From my dashboard analysis, I identified the following findings:

- Total revenue is approximately 4.71bn and total profit is approximately 979.89M.
- The dashboard records 15K orders and approximately 112K units sold.
- Mobile Phones is the highest-revenue category, followed by Laptops.
- Online stores contribute approximately 69.02% of total revenue, making this the largest store-type contribution.
- Mobile Phones and Laptops also occupy the largest areas in the profit treemap.
- Mumbai, New Delhi and Pune have the highest visible city revenue bars.
- Online Store has a substantially longer revenue bar than the other individual stores shown.
- The monthly revenue chart declines across its displayed axis sequence, although the month-year labels are not arranged as a simple calendar sequence.

10. Importance of the Dashboard

- **Sales Monitoring:** KPI cards provide a quick summary of revenue, profit, orders and quantity.
- **Category Analysis:** The column chart and treemap help me compare category-level revenue and profit.
- **Store-Type Analysis:** The doughnut chart shows the revenue contribution of each store format.
- **Time-Based Review:** The revenue line chart helps me inspect changes across the displayed month-year combinations.
- **Geographical Analysis:** The city chart helps me compare revenue across locations.
- **Store-Level Review:** The bar chart and table provide more detail about individual store performance.
- **Interactive Filtering:** The slicers allow me to examine selected years, store types, categories, customer segments and states.

11. Conclusion

Through this project, I learned how to use Microsoft Power BI to analyse retail sales and financial performance across products, store formats, cities and individual stores. I created KPI cards to summarize revenue, profit, orders and quantity, and used charts to compare revenue and profit across product categories and store types.

The city and store visuals helped me examine geographical and individual-store performance, while the year, store type, product category, customer segment and state slicers made the dashboard interactive. I also created a Store Type Tooltip page to present key revenue and profit measures together with order count and average order value.

Overall, this project improved my understanding of retail performance analysis, DAX measure validation, category comparisons, store-level reporting, interactive filters and tooltip design in Power BI.

12. References

- Microsoft Power BI Desktop.
- Retail sales dataset used for the project.
- Microsoft Power BI documentation.